PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Dose-response Effect of Postprocedural Elevated Cardiac Troponin Level on Adverse Clinical Outcomes Following Adult Noncardiac Surgery: A Systematic Review Protocol of Prospective Studies
AUTHORS	An, Tao; Tian, Yue; Guo, Jingfei; Kang, Wenying; Tian, Tao; Zhou, Chenghui

VERSION 1 – REVIEW

REVIEWER	Szczeklik, Wojciech Jagiellonian University Medical College, Department of Intensive Care and Perioperative Medicine
REVIEW RETURNED	22-Dec-2020

GENERAL COMMENTS	Thank you for the possibility to review: “Potential Dose-response Effect of Postprocedural Elevated Cardiac Troponin Level on Adverse Clinical Outcomes Following Noncardiac Surgery: A Systematic Review Protocol of Prospective Studies” by An Tao et. al. This is a protocol, following the PRISMA guidelines for a systematic review and meta-analysis planned to evaluate possible dose response relationship between postoperative elevated cardiac troponins (cTn) levels and adverse clinical outcomes after noncardiac surgery. 1. It is an interesting and important topic. It was already described in large studies (e.g VISION study PJ. Devereaux JAMA. 2012 6;307(21):2295-304) that the higher the troponin levels the higher the frequency of complications. Therefore, I am not fully supportive for a authors’ statement in the abstract: “effect of postprocedural cardiac troponin (cTn) levels on adverse clinical outcomes has not been studied’. This should be well discussed the authors as it gives good rationale for the proposed metanalysis (and probably a good hypothesis – the higher the level of troponins the more common the complications). 2. Please state clearly the definition that will be used for MINS – it differs from the acute myocardial injury definition (universal MI definition) and from the recent Standardized Endpoints in Perioperative Medicine (StEP) initiative document on cardiovascular outcomes (BJA 2020). 3. The authors could consider adding Chinese language for the search (as I understang the language of the authors)

	4. Sensitive analysis for age response and type of surgery could be added to the analysis 5. Authors should also consider sensitive analysis for different types of troponins. It is extremely difficult to compare high sensitive troponins to non-high sensitive Tn. Comparing different types of troponin I and troponin T in the same analysis may impose bias (e.g. elevation by 2 x TnT may not be the same as the same elevation for different type of TnI) 6. Discussion should be rewritten. The first paragraph is difficult to understand and consist several typos and not understandable sentences. I am not sure also why the authors should also limit discussing cardiac surgery, as the planned study is in noncardiac surgery: Examples: 'Although there has been several meta-analyses concerning about the perioperative troponin' 'Moreover, no previous meta-analysis has studied the potential linear or non-linear dose-response relationship between postoperative troponin level and adverse clinical outcomes in adult cardiac surgery'

REVIEWER	Weissman , Charles Hebrew University-Hadassah School of Medicine, Anesthesiology and Critical Care Medicine
REVIEW RETURNED	05-Jan-2021

GENERAL COMMENTS	This protocol proposes to perform a meta-analysis of studies that examined the prognostic abilities of cardiac troponin (cTn) levels on adverse clinical outcomes after non-cardiac surgery. This is an important topic since many studies have indicated that an elevated post-operative troponin concentration is associated with post-operative complications and poor outcome. 1. The authors need to take into account that elevated troponin has been observed in non-cardiac situations such as pulmonary embolism and thus might not be a marker of only direct cardiac problems but also indirect effects. 2. The effect of renal dysfunction on troponin concentrations should to be taken into account. 3. The type of surgery needs to be examined since the degree of hemodynamic and metabolic stress vary and might influence prognosis and complication rates. Similarly, age and sex are also important variables that should be examined, if possible. 4. In some of the studies, preoperative testing for cardiac ischemia was performed, these data should also be included in the analysis. 5. What will be the range of the literature review, will studies performed before 2000 be included? Specific comments 1. Introduction: "high-risk patients undergoing non-cardiac surgery" - please define "high-risk". 2. "The studies unable to extract odds ratio (OR) or hazard ratio (HR) and the corresponding 95% confidence intervals (CI) will be excluded." However, these studies should be scrutinized and any relevant information included in a narrative review section of the final report.

REVIEWER	Schälte, Gereon University Hospital Aachen, Dept. of Anesthesiology

REVIEW RETURNED	22-Jan-2021

GENERAL COMMENTS	A ambitious approach to study the clinical impact of postoperatively elevated troponin levels on patients' short and long term outcome.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Comments to the Author
Major
1. It is an interesting and important topic. It was already described in large studies (e.g VISION study PJ. Devereaux JAMA. 2012 6;307(21):2295-304) that the higher the troponin levels the higher the frequency of complications.
Therefore, I am not fully supportive for a authors' statement in the abstract:
"effect of postprocedural cardiac troponin (cTn) levels on adverse clinical outcomes has not been studied'.
This should be well discussed the authors as it gives good rationale for the
proposed metanalysis (and probably a good hypothesis – the higher the level of troponins the more common the complications).
Re: Thank you for the suggestion. We have revised it as follows: "However, whether higher level of postoperative cardiac troponin (cTn) possess higher incidence of major complications remains controversial.".

2. Please state clearly the definition that will be used for MINS – it differs from the acute myocardial injury definition (universal MI definition) and from the recent Standardized Endpoints in Perioperative Medicine (StEP) initiative document on cardiovascular outcomes (BJA 2020).
Re: Thank you for the constructive suggestion. We have added the related content in the METHODS AND ANALYSIS Section:
"Definition of MINS
We will not set a constant value for the definition of MINS. If a precise cutoff value has been provided in each study, the definition of MINS will be accepted. Three types of cutoff value will be existed: ① detection limit below URL; ② URL; ③ a value above URL. This definition is not according to the UDMI[6] or Standardized Endpoints in Perioperative Medicine initiative[19] for the exploration of dose-response relationship. "

3. The authors could consider adding Chinese language for the search (as I understang the language of the authors)
Re: Thank you for the advice. We have added it for the search in the Search strategy Section.

4. Sensitive analysis for age response and type of surgery could be added to the analysis
Re: Thank you for the advice. We have added it in the Data synthesis Section.

5. Authors should also consider sensitive analysis for different types of troponins. It is extremely difficult to compare high sensitive troponins to non-high sensitive Tn. Comparing different types of troponin I and troponin T in the same analysis may impose bias (e.g. elevation by 2 x TnT may not be the same as the same elevation for different type of TnI)
Re: Thank you for the advice. We have added it in the Data synthesis Section.

6. Discussion should be rewritten. The first paragraph is difficult to understand and consist several typos and not understandable sentences.
I am not sure also why the authors should also limit discussing cardiac surgery, as the planned study is in noncardiac surgery:
Examples: ‘Although there has been several meta-analyses concerning about the perioperative troponin’
‘Moreover, no previous meta-analysis has studied the potential linear or non-linear dose-response relationship between postoperative troponin level and adverse clinical outcomes in adult cardiac surgery’
Re: Sorry for the wrong expressions. We have corrected the “cardiac surgery” as “noncardiac surgery”. Additionally, we have largely rewritten the first paragraph of Discussion Section.

Special thanks for your constructive advices.

Reviewer: 2
Comments to the Author
Major

1. The authors need to take into account that elevated troponin has been observed in non-cardiac situations such as pulmonary embolism and thus might not be a marker of only direct cardiac problems but also indirect effects.

Re: Thank you for your suggestion. We have stated this issue as a limitation.

2. The effect of renal dysfunction on troponin concentrations should to be taken into account.

Re: Thank you for your advice. We have stated this issue as a limitation similar with pulmonary embolism.

3. The type of surgery needs to be examined since the degree of hemodynamic and metabolic stress vary and might influence prognosis and complication rates. Similarly, age and sex are also important variables that should be examined, if possible.

Re: Thank you for the advice. We have added it in the Data synthesis Section.

4. In some of the studies, preoperative testing for cardiac ischemia was performed, these data should also be included in the analysis.

Re: Thank you for the advice. We have added it in the Data synthesis Section.

5. What will be the range of the literature review, will studies performed before 2000 be included?

Re: Thank you for your concerns. We will not limit the publication year for the inclusion of studies. There is only one eligible study before 2000 (J Am Coll Cardiol . 1997 May;29(6):1241-5.).

Specific comments

1. Introduction: "high-risk patients undergoing non-cardiac surgery" - please define "high-risk".

Re: Thank you for your concerns. We have defined the related content in the Introduction Section.

2. "The studies unable to extract odds ratio (OR) or hazard ratio (HR) and the corresponding 95% confidence intervals (CI) will be excluded." However, these studies should be scrutinized and any relevant information included in a narrative review section of the final report.

Re: Thank you for your carefulness. We have revised it accordingly.

Thank you for your comments and suggestions.

Reviewer: 3
Comments to the Author
Minor
A ambitious approach to study the clinical impact of postoperatively elevated troponin levels on patients' short and long term outcome.
Re: Thank you for your good comment.

Special thanks for your kind advice!

We tried our best to improve the manuscript and made some corrections in the manuscript according to the Editorial and Reviewers' comments. These changes will not affect the content and framework of the paper. We did not list the minor changes in the response letter. We appreciate for Editors/Reviewers' warm work earnestly, and hope that the corrections will meet with approval.
Once again, thank you very much for your comments and suggestions.

VERSION 2 – REVIEW

REVIEWER	Szczeklik, Wojciech Jagiellonian University Medical College, Department of Intensive Care and Perioperative Medicine
REVIEW RETURNED	07-Mar-2021

GENERAL COMMENTS	Thank you for addressing my comments.

REVIEWER	Weissman , Charles Hebrew University-Hadassah School of Medicine, Anesthesiology and Critical Care Medicine
REVIEW RETURNED	25-Feb-2021

GENERAL COMMENTS	This manuscript has undergone revision, but there are a number of remaining issues. 1. The manuscript needs extensive English language editing. 2. The "Strengths and limitations of this study" Section is unclear "This meta-analysis will include the largest prospective study population" – please explain "largest prospective study" Do you mean you will pool data from a number of studies to form the largest database of prospective data? "The baseline cTn level is not a regular test for patients undergoing non-cardiac surgery" – do you mean "routine" test? 3. Page 6: "For the studies unable to extract odds ratio (OR) or hazard ratio (HR) and the corresponding 95% confidence intervals (CI), we will perform a related systematic review in a section" – will this be a systematic or narrative review? 4. The "Definition of MINS". This section is unclear and needs editing. Also, justify why "This definition is not according to the UDMI6 or Standardized Endpoints in Perioperative Medicine initiative". 5. The authors will be using multiple studies of non-cardiac surgery. How do they expect to control for the type of surgery and the different patient characteristics?

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Comments to the Author:

1. The manuscript needs extensive English language editing.

Re: Thank you for the advice. We have completed the proofread process via AJE.

2. The "Strengths and limitations of this study" Section is unclear

"This meta-analysis will include the largest prospective study population" – please explain "largest prospective study" Do you mean you will pool data from a number of studies to form the largest database of prospective data?

Re: Sorry for the unclear expression. We have corrected it accordingly.

3. "The baseline cTn level is not a regular test for patients undergoing non-cardiac surgery" – do you mean "routine" test?

Re: Sorry for the inappropriate expression. We have corrected it accordingly.

4. Page 6: "For the studies unable to extract odds ratio (OR) or hazard ratio (HR) and the corresponding 95% confidence intervals (CI), we will perform a related systematic review in a section" – will this be a systematic or narrative review?

Re: Sorry for the misleading statement. We will not include the studies unable to extract OR or HR, and have deleted the related phrase.

5. The "Definition of MINS". This section is unclear and needs editing. Also, justify why "This definition is not according to the UDMI6 or Standardized Endpoints in Perioperative Medicine initiative".

Re: Thank you for your constructive advice. We have corrected it as following: “The definition of MINS with a precise cut-off value in each study will be accepted. The following three types of cut-off value will exist: ① detection limit below the URL; ② detection at the URL; ③ detection above the URL.This definition based only on biomarkers of myocardial injury is not based on the UDMI[6] or Standardized Endpoints in Perioperative Medicine initiative[19] due to the lack of availability of additional information such as electrocardiography, echocardiography, coronary CT or angiography data .”

6. The authors will be using multiple studies of non-cardiac surgery. How do they expect to control for the type of surgery and the different patient characteristics?

Re: Thank you for your advice. We have revised the Data synthesis in Methods and Analysis section:” Univariable or multivariable meta-regression and subgroup analyses will be conducted for the comparison between patients with elevated versus nonelevated postoperative cTn levels to

assess the impact of multiple potential influential factors such assurgical types, patient characteristics, and cTn types (high sensitive versus non-high sensitive, cTnI versus cTnT, baseline cTn versus without baseline cTn)" .

Special thanks for your constructive advices.

We tried our best to improve the manuscript and made some corrections in the manuscript according to the Editorial and Reviewers' comments. These changes will not affect the content and framework of the paper. We did not list the minor changes in the response letter. We appreciate for Editors/Reviewers' warm work earnestly, and hope that the corrections will meet with approval.

Once again, thank you very much for your comments and suggestions.

VERSION 3 – REVIEW

REVIEWER	Weissman , Charles Hebrew University-Hadassah School of Medicine, Anesthesiology and Critical Care Medicine
REVIEW RETURNED	31-May-2021

GENERAL COMMENTS	Adequate revision